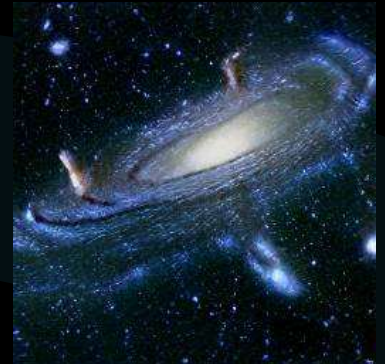
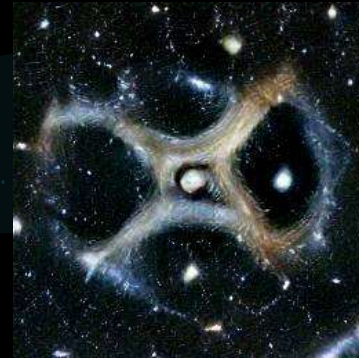


Welcome to the Numerai Universe

A one year journey by QE



Who is QE??

- Dutchie, 44, girlfriend and son, cycling, table tennis
- IT Consultant (cloud / software / data solutions)
- RPI / Arduino robot who likes the cloud and Numerai
- Data Scientist Wannabe



★ Looking for a challenge...

- Study is nice, but practice is better!



Microsoft Professional Program
Artificial Intelligence Certificate

- Which tournament to pick??



- ★ The most difficult one!!



Start of the journey, 03-2021

Numerai payouts



numerai-cli 0.2



Numerati



- Number of Classic Stakes: 2000, 240.000 NMR
- Number of Classic Models : ~6000
- V2 Data: 310 Features, 1 target, 142 eras



GraphQL

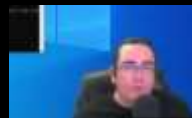
tournament api



numerapi 2.4.5



NUMERAI SIGNALS



What is it about??



Training? Validation? Models?



- Curious about NN
- Curious about 'classic' algorithms
- Curious about AutoML
- Try to avoid overfitting
- Try not to lose money!



TensorFlow



Keras

XGBoost



End Result?



- *Best Corr ranking: 81 (not for long)*
- *Best TC ranking: 2 (not for long)*
- *Earned almost no coins*
- *Learned a Lot!!*



- NN works for a limited time
 - Model Kakkie (V2!)
- AutoML can give quite good results with proper configuration
 - Models Villager, xiao_zhishi_2, 3
- Small/medium feature-set is a good start
- Split up that data! -> overfitting
- Try different training data!
 - Validation = Training

Give me some Insights!



tournament api



ORACLE®

Cloud Infrastructure



Classics Dashboard

Compute?

local



Cloud



Numerai compute
(~1-xx dollars)



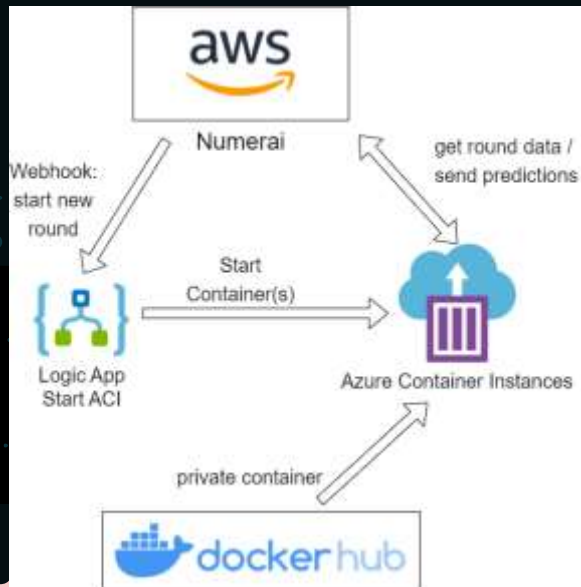
prediction node



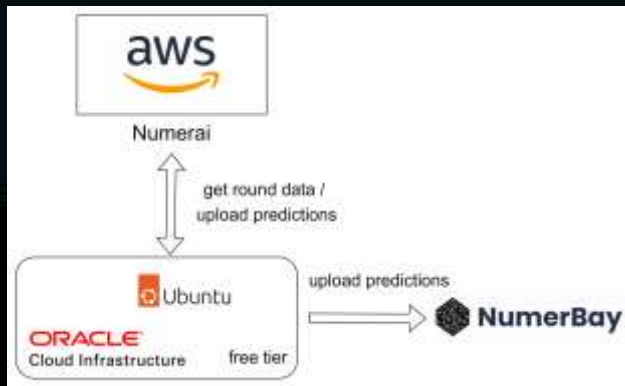
FLOYDHUB

Compute!!

Azure (~1-xx dollars)



Oracle (~0-xx dollars)



Always Free cloud services

Services you can use for an unlimited time:

- Two Oracle Autonomous Databases with powerful tools like Oracle APEX and Oracle SQL Developer
- Two AMD Compute VMs
- Up to 4 instances of ARM Ampere A1 Compute with 3,000 OCPU hours and 18,000 GB hours per month

- Own server -> Flexibility
- 24x7 -> Daily Compute
- 2x x86/x64, 1Gb RAM, Swap 12-16Gb
- 1x Arm64 (Ampere A1), 24Gb RAM
- 40 models (TF, regression) < 30 mins

https://github.com/jos1977/numerair_compute

Oracle AMD

Cronjob

orchestration

Download
from Numerai

Load live
data in memory

Predict/upload
on all models

Upload
to Numberbay

```
#example_model.py, this should be customized and extended to your own model(s)
# if run example_model_submitted:

PREDICTION_FILE = "/predictions/prediction_examplemodel_v4." + str(current_round) + ".csv"
PREVIOUS_PREDICTION_FILE = "/predictions/prediction_examplemodel_v4." + str(previous_round)
MODEL_FILE = "/models/examplemodel_v4/model.pkl"

logging.info("PREDICTION FILE is: %s", PREDICTION_FILE)
logging.info("PREVIOUS PREDICTION FILE is: %s", PREVIOUS_PREDICTION_FILE)
logging.info("MODEL FILE is: %s", MODEL_FILE)

if os.path.isfile(PREVIOUS_PREDICTION_FILE):
    os.remove(PREVIOUS_PREDICTION_FILE)
    #writing the confirmation message of deletion
    print("File deleted successfully")
else:
    print("File does not exist")

model = joblib.load(MODEL_FILE)

live_data["preds"] = model.predict(live_data[features])

live_data["preds_neutral_riskiest_50"] = neutralize(
    live_data,
    columns=["preds"],
    neutralize=richest_features,
    proportions=0,
    normalize=True,
    era_tol=0.05)

# remove best model by "prediction" and rank from 0 to 1 to most optimal requirements
live_data["prediction"] = live_data["preds_neutral_riskiest_50"].rank(pct=True)
```

The screenshot shows the Oracle Compute console interface. At the top, there's a navigation bar with 'main' and 'numeraicompute / oracle'. Below that, a table lists various nodes with columns for 'nodes' and 'Oracle example'. The nodes listed include 'PREDICTing', 'numera_requirements', 'compute_neutralize', 'offcpu', 'v1_200gb', 'v1_200gb', 'v1_200gb', and 'v1_200gb'. Below the table, there's a 'README' section titled 'Oracle Compute' which contains a disclaimer and installation steps. The disclaimer states that the example code and instructions are provided as-is and that users should check their cloud costs. The installation steps mention creating an Oracle Free Tier Account and Compute Engine.

nodes	Oracle example
PREDICTing	compute-example
numera_requirements	Oracle-example
compute_neutralize	Oracle-example
offcpu	Oracle-example
v1_200gb	Oracle-example
v1_200gb	Oracle-example
v1_200gb	Oracle-example
v1_200gb	Oracle-example

Oracle Compute

This repository contains the numeraicompute example in combination with Oracle Cloud provider. The examples here are meant to be used for benchmarking predictions. The Oracle compute version is based on the free tier and at last doesn't incur any cost. There are two compute engines available: amd64 (aws) and amd64 (aws) with 1GB ram. The following example will be based on AMD but could also potentially work for the aws version (depending on library requirements).

Disclaimer

No guarantees are given that the example code and instructions here are 100% correct, and make sure to keep track of your cloud costs, this code may with some little mistakes to increase costs considerably. As such, if these examples are used, you use them at your own risk.

Installation steps

For more the installation steps are manual and require knowledge about server maintenance, ssh/ftp and basic linux (ubuntu) knowledge. Performs the following steps to create the Ubuntu server and get the server configured for daily/weekly predictions at Numerai.

Create an Oracle Free Tier Account and Compute Engine

The cloud provider used is Oracle Infrastructure which also a free tier and is in principle sufficient for any basic numerai model (regression/ML).

Welcome to the Numerai Universe



NumerBay!



Numerai payouts

Awesome Numerai



Numerati



numerai-cli 0.3.4



GraphQL

tournament api



numerapi 2.12.1

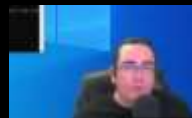
- Number of Classic Stakes: 4400, 700.000 NMR
- Number of Classic Models : ~15000
- V2 Data: 310 Features, 1 target, 142 eras
- V3 Data: 1050 Features, 1 target, 679 eras
- V4 Data: 1194 Features, >10 targets, >679 eras



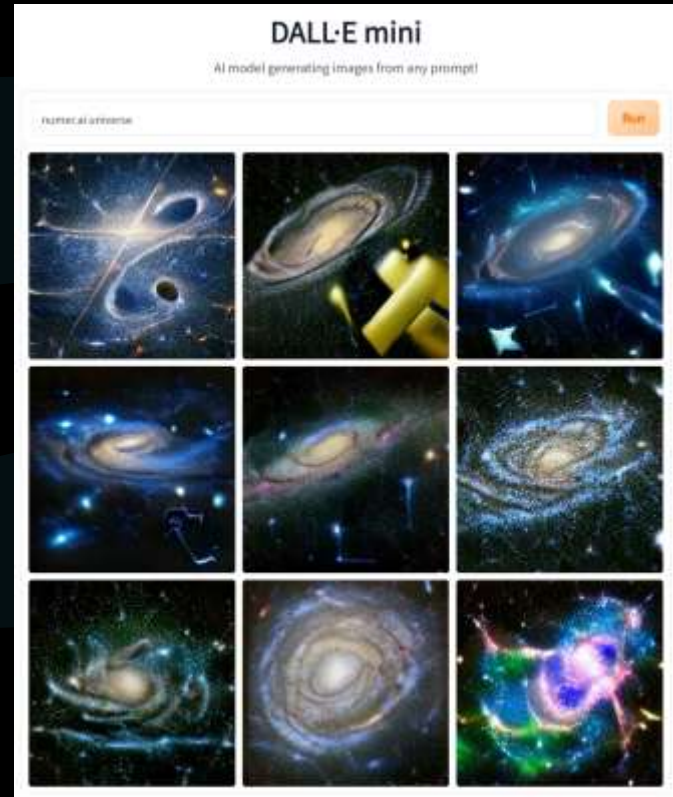
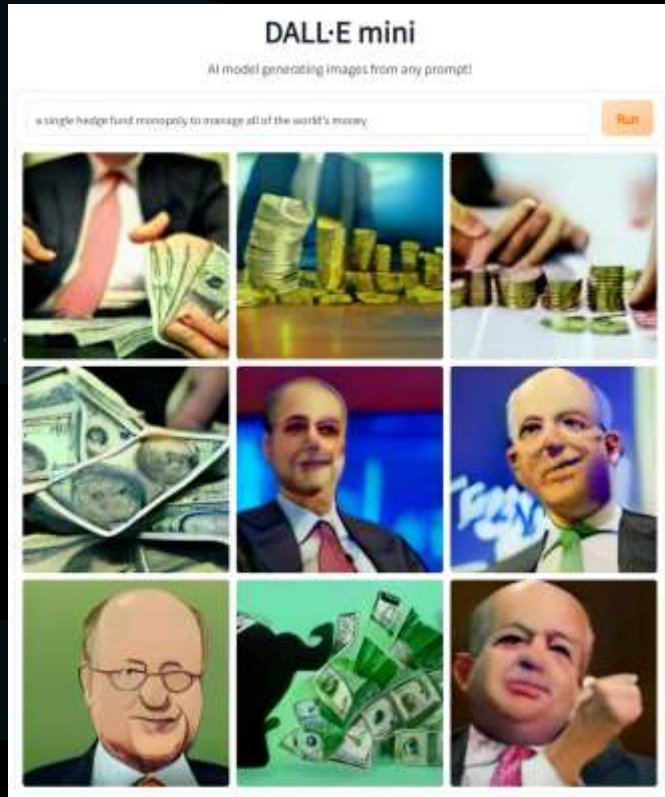
numberblox



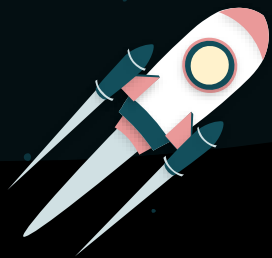
open signals



Fun with Dall E mini



THANKS!



Any questions?

★ All the links

Apps / Platforms

- <https://numerbay.ai/>
- <https://github.com/crowdcent/numerblox>

API's

- <https://github.com/numeraai/numeraai-cli>
- <https://github.com/Omni-Analytics-Group/Rnumeraai>
- <https://api-tournament.numer.ai/>
- <https://github.com/uuazed/numerapi>

✧ All the links

Dashboards

- <https://dashboard.numerai.payouts.com/>
- <https://app.powerbi.com/view?r=eyJrIjoieTYiOTk2ZjYtYTIiYy00ZjQ5LWE2NjMtMTBjOTkwYTg2NDE2liwidCI6Ijg3ZDc2ZDQ2LWYxZmYtNDkzMj05MGNiLTUyNzY3Yzg2OTk2ZjZlslmMiOjI9>
- https://github.com/jos1977/numerai_statistics
- <https://www.jofaichow.co.uk/numerati/data.html>
- <https://www.numerai-insights.com/>
- https://share.streamlit.io/yifanxie/numerdash/main/numerdash_app.py
- <https://dune.com/aventurine/NumeraiNMR>
- <https://numerai-earnings.herokuapp.com/>

Other

- <https://github.com/councilofelders>
- <https://github.com/peterling7710/NumeraiStarterPack>
- https://github.com/jos1977/numerai_compute
- <https://gosuto.medium.com/kashi-nmr-borrowing-stake-on-your-model-without-nmr-price-exposure-bc687fcacd9f>
- <https://gosuto.medium.com/kashi-nmr-lending-earn-apr-without-staking-on-a-model-108c49bc464d>
- <https://coenumerainewsletter.substack.com/>